

CHALLENGE TRACK 3 · HAMILTONIAN SIMULATION

Watching a Quantum Magnet Evolve in Time

From a first Trotter step to trustworthy dynamics at the edge of classical reach.

Background and Motivation

Simulating how a quantum system evolves in time was Feynman's original argument for building a quantum computer. Given a Hamiltonian $\hat{H}$, a state evolves under the time-evolution operator $U(t) = e^{-i\hat{H}t}$ — but for a many-body system this operator is exponentially large and cannot generally be compiled exactly. **Trotterization** (a product formula) sidesteps this by slicing time into small steps and applying the easy pieces of the Hamiltonian one after another, turning an intractable exponential into a repeating pattern of elementary gates.

Our testbed is the **transverse-field Ising model (TFIM)**, a cornerstone of quantum magnetism, whose competing interaction and field terms produce rich dynamics. Digital simulation of exactly this kind of spin-chain evolution on IBM hardware (Smith et al., 2019) inspires the entry point of this track; large-scale, error-mitigated kicked-Ising simulation (Kim et al., 2023) inspires its summit. A decade into cloud quantum computing, this challenge asks you to make dynamics both accurate and trustworthy.

> **The scientific question**
>
> How faithfully can a digital quantum computer reproduce the time evolution of a quantum magnet — and how do we keep the answer trustworthy as circuits deepen and classical verification fades?

The model

Throughout this track we work with the one-dimensional transverse-field Ising model on N spins with open boundary conditions,

$$\hat{H}_{\mathrm{TFIM}} = -J \sum_{i=0}^{N-2} Z_i Z_{i+1} \; - \; h \sum_{i=0}^{N-1} X_i, \tag{1}$$

where Z_i and X_i are Pauli operators on site i, J is the nearest-neighbour coupling, and h is the transverse-field strength. Unless a level states otherwise, take $J = 1$ and $h = 1$ (the interaction and field are comparable, giving non-trivial dynamics), and start from the fully polarized state $|\psi(0)\rangle = |00\cdots0\rangle$.

The primary observable is the average magnetization along z,

$$M_z(t) = \frac{1}{N} \sum_{i=0}^{N-1} \langle\psi(t)|\, Z_i \,|\psi(t)\rangle\,, \tag{2}$$

and, where correlations are needed, the two-point function $C_{ij}(t) = \langle Z_i Z_j \rangle - \langle Z_i \rangle \langle Z_j \rangle$.

The companion tutorial notebook builds the first-order Trotterized evolution of Eq. (1) and tracks Eq. (2) for a small chain, checked against exact evolution. The challenge levels below take this further.

Getting Started — Beginner

Your target

Take the TFIM of Eq. (1) with $N \in \{4, 5, 6\}$, $J = h = 1$, initial state $|00\cdots0\rangle$, and first-order (Lie–Trotter) time evolution. Produce a study of its Trotterized dynamics that delivers all three of:

(a) a two-point correlation function $C_{ij}(t)$ tracked over time;
(b) the dependence of Trotter error on the step size Δt, for a fixed final time T;
(c) a comparison of first-order Lie–Trotter against a higher-order (Suzuki–Trotter) formula at matched accuracy.

Every quantitative claim should be supported by a plot benchmarked against exact evolution.

Suggested milestones

These are optional pointers, not requirements — how you reach the target is up to you.

- Compute $C_{ij}(t)$ as a function of separation $|i - j|$ and time; ask how quickly correlations spread along the chain.
- For a fixed T, sweep Δt and check whether you can see the expected first-order error scaling.
- Plot error against circuit depth (or two-qubit gate count) to expose the accuracy-versus-depth trade-off, and find a sweet spot.
- For a fixed accuracy target, compare the *total* gate count of first- and second-order formulas.

Next Steps — Intermediate and Advanced

The two levels below are open-ended: the target defines *what* to achieve, and the milestones merely suggest *how*.

Intermediate — Recover Physics the Noise Is Hiding

Your target

Simulate the dynamics of a larger instance of Eq. (1) — a chain of $N \approx 10$–12 spins, or a small two-dimensional lattice of comparable size — under realistic noise (a noisy simulator or real hardware). Using error mitigation, recover a magnetization $M_z(t)$ or correlation $C_{ij}(t)$ curve that the *unmitigated* device gets visibly wrong, and present the mitigated result against a trusted reference.

Suggested milestones

- Apply error mitigation such as zero-noise extrapolation, measurement-error mitigation, or Pauli twirling.
- Use circuit-compression or transpilation tricks that lower depth without changing the physics.

- Show, side by side, the noisy, mitigated, and reference curves.

Advanced — Simulate Where Classical Checking Gets Hard

For the summit we move to a periodically driven (Floquet) system. In the **kicked Ising model**, evolution proceeds in discrete cycles, each applying an Ising interaction followed by a global transverse "kick":

$$U_{\text{cycle}} = \underbrace{\exp\Big(i\,\theta_h \sum_i X_i\Big)}_{\text{transverse kick}} \underbrace{\exp\Big(i\,\theta_J \sum_i Z_i Z_{i+1}\Big)}_{\text{Ising interaction}}, \qquad |\psi(n)\rangle = U_{\text{cycle}}^{n}\,|\psi(0)\rangle\,. \tag{3}$$

The state after n cycles is $|\psi(n)\rangle$; a natural observable is the average magnetization $M_z(n)$ as a function of the number of kicks.

Your target

Simulate the kicked Ising evolution of Eq. (3) on $N \gtrsim 20$ spins on real hardware — a scale where brute-force exact classical simulation is no longer practical — and use error mitigation to produce a *trustworthy* expectation value (for example $M_z(n)$ after a fixed number of kicks). Defend the correctness of your result *without* relying on exact diagonalization.

Suggested milestones

- Choose kick angles (θ_J, θ_h) and justify the regime you study (e.g. near a known limit).
- Lean on lightweight classical checks, known limiting cases (e.g. $\theta_h \to 0$ or $\pi/2$), and consistency across mitigation settings.
- Report the largest N / longest evolution whose result you can defend, and explain the argument.

Running on real hardware (all levels)

On a real device (such as `ibm_basquecountry`), the heavy-hex connectivity means the ZZ interactions in Eqs. (1)–(3) may not map onto physically connected qubits, forcing SWAPs that deepen the circuit. Mapping your chain onto a well-connected line of qubits, keeping Trotter steps shallow, and applying error suppression / mitigation are all part of a good design.

Suggested Resources

Trotterization and time evolution in Qiskit

Quantum real-time evolution using Trotterization
Simulating dynamics with product formulas, including Lie–Trotter, Suzuki–Trotter, and depth analysis.

PauliEvolutionGate documentation
Reference for the gate representing $e^{-i\hat{H}t}$ and how the product-formula synthesis is chosen.

LieTrotter and SuzukiTrotter synthesis
First-order and higher-order product formulas — directly relevant to the beginner target.

TrotterQRTE documentation
A higher-level driver that automates Trotterized evolution and evaluates observables at each step.

Observables, primitives, and operators

SparsePauliOp documentation
Building Hamiltonians and observables as weighted sums of Pauli strings.

Introduction to the primitives
The Estimator and Sampler primitives and how they run on simulators and hardware.

Hardware, transpilation, and error mitigation

Transpile with pass managers
Transpiling circuits for a backend and its heavy-hex connectivity.

Error mitigation and suppression techniques
Overview of resilience options — central to the intermediate and advanced levels.

Background reading

Smith et al., Simulating quantum many-body dynamics on a current digital quantum computer, npj QI (2019)
Digital simulation of spin-chain dynamics on IBM hardware — inspiration for the beginner track.

Kim et al., Evidence for the utility of quantum computing before fault tolerance, Nature (2023)
Large-scale kicked-Ising simulation with error mitigation — inspiration for the advanced track.